

Vinogradov–Korobov Growing-Order Prime-Tail Uniformization

RH research program

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Abstract

Let $3 = p_1 < p_2 < \dots$ be the odd primes, put $x = p_y$ and $L = \log x$, and define the strict tails

$$P_r(y) = \sum_{p > x} (p^2 - 1)^{-r} \quad (r \geq 1).$$

Using the explicit Vinogradov–Korobov estimate of Johnston and Yang, we prove a quantitative comparison of P_r with three successive kernels. With

$$V = L^{3/5}(\log L)^{-1/5}, \quad \varepsilon_x = 0.027L^{1.801}e^{-0.1853V},$$

the exact Stieltjes kernel J_r , power kernel I_{2r} , and leading term $K_r = x^{1-2r}/((2r-1)L)$ satisfy

$$\left| \log \frac{P_r}{J_r} \right| \leq 14r\varepsilon_x, \quad 0 \leq \log \frac{J_r}{I_{2r}} \leq \frac{r}{x^2 - 1}, \quad \left| \log \frac{I_{2r}}{K_r} \right| \leq \frac{1}{(2r-1)L},$$

whenever $L \geq 512$ and $7r\varepsilon_x \leq 1/2$. For a partition $\lambda = 1^{k_1}2^{k_2}\dots$, set $d = \sum rk_r$, $H = \sum k_r/(2r-1)$, and $H_2 = \sum k_r/(2r-1)^2$. The corresponding product has the refined uniform expansion

$$\log \frac{\mathcal{P}_\lambda}{\mathcal{M}_\lambda} = -\frac{H}{L} + O\left(d\varepsilon_x + \frac{d}{x^2 - 1} + \frac{H_2}{L^2}\right),$$

with explicit constants 14, 1, 2. Thus $d\varepsilon_x + d/x^2 \rightarrow 0$ is a robust source-and-power condition, while the leading equivalent holds precisely when $H/L \rightarrow 0$. In particular $\log d = o(V)$ and $H = o(L)$ suffice. For a single order the range $\log R = o(V)$ is uniform, and one may sharpen it to $R \leq e^{(0.1853-\delta)V}$ for fixed $0 < \delta < 0.1853$. The family $\lambda = 1^{\lfloor cL \rfloor}$ has leading ratio tending to e^{-c} , proving the H/L obstruction is real. An exact 96-row artifact reproduces the algebraic interfaces but is not an analytic proof.

Keywords: explicit prime number theorem; Vinogradov–Korobov estimate; strict Stieltjes boundary; prime-square tail; growing integer partitions; uniform asymptotics.

1 Scope and notation

The fixed-order asymptotic

$$P_r(y) \sim \frac{x^{1-2r}}{(2r-1)\log x}$$

follows from the prime number theorem when r is fixed. That statement does not by itself permit r , the degree of a partition, or the number of factors to grow with y . The purpose of this paper is to pay for those three kinds of growth explicitly. The new input is the global explicit Vinogradov–Korobov estimate of Johnston and Yang [1, Theorem 1.4, equation (1.8)]; the prior fixed-order prime-tail calculation in RH-384 is used only as a finite regression target for the accompanying artifact [2].

Throughout,

$$3 = p_1 < p_2 < \cdots, \quad x = p_y, \quad L = \log x,$$

and r is a positive integer. Write

$$\vartheta(t) = \sum_{p \leq t} \log p, \quad E(t) = \vartheta(t) - t.$$

The strict prime tail and its three comparison kernels are

$$P_r(y) = \sum_{p > x} \frac{1}{(p^2 - 1)^r}, \tag{1}$$

$$h_r(t) = \frac{(t^2 - 1)^{-r}}{\log t}, \quad J_r(x) = \int_x^\infty h_r(t) dt, \tag{2}$$

$$I_{2r}(x) = \int_x^\infty \frac{t^{-2r}}{\log t} dt, \quad K_r(x) = \frac{x^{1-2r}}{(2r-1)L}. \tag{3}$$

The endpoint in (1) is always strict. In particular,

$$P_r(y) = \frac{1}{(p_{y+1}^2 - 1)^r} + P_r(y+1). \tag{4}$$

This exact successor identity distinguishes $p > x$ from $p \geq x$ even though their leading asymptotics agree.

Set

$$V(L) = L^{3/5}(\log L)^{-1/5}, \quad \varepsilon_x = \frac{27}{1000} L^{1801/1000} \exp\left(-\frac{1853}{10000} V(L)\right). \tag{5}$$

Johnston and Yang prove that, for every $t \geq 23$,

$$|E(t)| \leq t\varepsilon_t. \tag{6}$$

Their Corollary 1.2, equation (1.5), and the $X = \log 2$ row of their Table 1 are retained in the external source lock as provenance locators, but the derived square-root-exponential fallback is not used or asserted as a result here. Only (6) enters the proof.

Proposition 1.1 (Monotone source envelope). *The function of L on the right of (5) is strictly decreasing for $L \geq 512$.*

Proof. Logarithmic differentiation gives

$$\frac{d}{dL} \log \varepsilon = \frac{1}{L} \left[\frac{1801}{1000} - \frac{1853}{10000} V(L) \left(\frac{3}{5} - \frac{1}{5 \log L} \right) \right].$$

For $L \geq 512$, one has $\log L > 6$, $\sqrt{L} > 22$, and $V(L) \geq \sqrt{L}$. The last inequality is equivalent to $\sqrt{L} \geq \log L$, which holds on this range. Hence the factor in parentheses is at least $17/30$, and

$$\frac{1853}{10000} 22 \frac{17}{30} - \frac{1801}{1000} > 0.$$

The displayed derivative is therefore negative. □

2 The strict Stieltjes transfer

The endpoint and the boundary term must be kept exact before applying the source estimate.

Lemma 2.1 (Strict Stieltjes identity). *For $x = p_y$ and $r \geq 1$,*

$$P_r(y) = \int_{(x, \infty)} h_r(t) d\vartheta(t) = -\vartheta(x)h_r(x) - \int_x^\infty \vartheta(t)h'_r(t) dt. \quad (7)$$

Consequently,

$$P_r(y) - J_r(x) = -E(x)h_r(x) - \int_x^\infty E(t)h'_r(t) dt. \quad (8)$$

Proof. At a prime p , the jump of ϑ is $\log p$, so the first integral equals (1). Stieltjes integration by parts on $(x, R]$ gives the second expression with the lower boundary $-\vartheta(x)h_r(x)$. The upper boundary tends to zero; for example $\vartheta(R) \leq R \log R$ and $h_r(R) = O(R^{-2r}/\log R)$. Substitute $\vartheta(t) = t + E(t)$ and use

$$-xh_r(x) - \int_x^\infty th'_r(t) dt = \int_x^\infty h_r(t) dt = J_r(x)$$

to obtain (8). □

Lemma 2.2 (Decreasing hazard). *The logarithmic hazard*

$$q_r(t) = -\frac{h'_r(t)}{h_r(t)} = \frac{2rt}{t^2 - 1} + \frac{1}{t \log t} \quad (9)$$

is positive and strictly decreasing for $t > 1$. Moreover,

$$J_r(x) \geq \frac{h_r(x)}{q_r(x)}, \quad xq_r(x) \leq 3r \quad (x \geq 23). \quad (10)$$

Proof. Both summands in (9) are positive and strictly decreasing: their derivatives have numerators $-2r(t^2 + 1)$ and $-(\log t + 1)$, respectively, after multiplication by positive denominators. Hence

$$h_r(x) = \int_x^\infty q_r(t)h_r(t) dt \leq q_r(x)J_r(x).$$

Finally

$$xq_r(x) = \frac{2rx^2}{x^2 - 1} + \frac{1}{\log x} \leq r \left(2 + \frac{2}{528} + \frac{1}{3} \right) < 3r$$

for $x \geq 23$. □

Proposition 2.3 (Explicit source-to-kernel transfer). *Assume $L \geq 512$. Then*

$$\left| \frac{P_r(y)}{J_r(x)} - 1 \right| \leq 7r\varepsilon_x. \quad (11)$$

If in addition $7r\varepsilon_x \leq 1/2$, then

$$\left| \log \frac{P_r(y)}{J_r(x)} \right| \leq 14r\varepsilon_x. \quad (12)$$

Proof. Since $h'_r < 0$, Proposition 1.1, (6), and (8) imply

$$\begin{aligned} |P_r - J_r| &\leq \varepsilon_x \left(xh_r(x) + \int_x^\infty t(-h'_r(t)) dt \right) \\ &= \varepsilon_x (2xh_r(x) + J_r(x)). \end{aligned}$$

Lemma 2.2 gives $xh_r(x)/J_r(x) \leq 3r$, so the relative error is at most $(6r + 1)\varepsilon_x \leq 7r\varepsilon_x$. If this is at most $1/2$, then $|\log(1 + u)| \leq 2|u|$ for $|u| \leq 1/2$, proving (12). □

Remark 2.4 (Why the boundary is not optional). The two copies of $xh_r(x)$ in the proof have different origins: one is the strict Stieltjes boundary and one comes from $\int t(-h'_r) = xh_r + J_r$. Deleting the boundary changes the transfer algebra. Conversely, the leading asymptotic alone cannot detect an inclusive-endpoint mutation, which is why (4) is also retained.

3 Two kernel comparisons

Lemma 3.1 (Exact kernel to power kernel). *For every $x > 1$ and $r \geq 1$,*

$$0 \leq \log \frac{J_r(x)}{I_{2r}(x)} \leq -r \log(1 - x^{-2}) \leq \frac{r}{x^2 - 1}. \quad (13)$$

No smallness assumption on r/x^2 is required.

Proof. The quotient J_r/I_{2r} is the weighted average, with positive weight $t^{-2r}/\log t$, of $(1-t^{-2})^{-r}$. That factor decreases from $(1-x^{-2})^{-r}$ to 1. This gives the first two inequalities. The last follows from $-\log(1-u) \leq u/(1-u)$ with $u = x^{-2}$. $\square$

The optional coarser estimate $\log(J_r/I_{2r}) \leq 4r/x^2$ used in an earlier finite interface follows immediately on the present range (and in particular under the formerly declared condition $r/x^2 \leq 3/8$). It is not used in the canonical ledger.

Lemma 3.2 (Power kernel to leading kernel). *Put*

$$a_r = \frac{1}{(2r-1)L}, \quad G(a) = \int_0^\infty \frac{e^{-v}}{1+av} dv.$$

Then

$$\frac{I_{2r}(x)}{K_r(x)} = G(a_r), \quad \frac{1}{1+a} \leq G(a) \leq 1, \quad (14)$$

and therefore

$$\left| \log \frac{I_{2r}(x)}{K_r(x)} \right| \leq a_r = \frac{1}{(2r-1)L}. \quad (15)$$

For $0 \leq a \leq 1/4$ one also has

$$0 \leq \log G(a) + a \leq 2a^2. \quad (16)$$

Proof. Set $t = xe^{v/(2r-1)}$ in I_{2r} . This gives the identity in (14). If V_0 is exponentially distributed with mean one, then $G(a) = \mathbb{E}(1 + aV_0)^{-1}$. Jensen's inequality and the trivial upper bound give $1/(1+a) \leq G(a) \leq 1$, hence $|\log G(a)| \leq \log(1+a) \leq a$.

For the refinement, $1/(1+a) \geq e^{-a}$ implies $\log G(a) + a \geq 0$. The pointwise inequality $(1+z)^{-1} \leq 1 - z + z^2$ and the exponential moments $\mathbb{E}V_0 = 1$, $\mathbb{E}V_0^2 = 2$ yield $G(a) \leq 1 - a + 2a^2$. Since $\log u \leq u - 1$, $\log G(a) + a \leq 2a^2$. $\square$

4 Growing partitions and the uniform theorem

Definition 4.1 (Partition statistics). Let $\lambda = 1^{k_1}2^{k_2}\dots$ be a nonempty integer partition, with only finitely many nonzero k_r . Define

$$\ell = \sum_r k_r, \quad d = \sum_r rk_r, \quad R = \max\{r : k_r > 0\}, \quad (17)$$

and

$$H = \sum_r \frac{k_r}{2r-1}, \quad H_2 = \sum_r \frac{k_r}{(2r-1)^2}. \quad (18)$$

Thus $0 < H_2 \leq H \leq d$ and $R \leq d$. For $A_r \in \{P_r, J_r, I_{2r}, K_r\}$ write

$$\mathcal{A}_\lambda = \prod_r A_r^{k_r},$$

using the corresponding calligraphic letter. In particular,

$$\mathcal{M}_\lambda(x) = x^{-(2d-\ell)} L^{-\ell} \prod_r (2r-1)^{-k_r}. \quad (19)$$

Theorem 4.2 (Quantitative growing-order uniformization). *Let $x = p_y$, $L \geq 512$, and let λ be a nonempty partition such that $7R\varepsilon_x \leq 1/2$. Then*

$$\left| \log \frac{\mathcal{P}_\lambda}{\mathcal{J}_\lambda} \right| \leq 14d\varepsilon_x, \quad (20)$$

$$\left| \log \frac{\mathcal{P}_\lambda}{\mathcal{I}_\lambda} \right| \leq 14d\varepsilon_x + \frac{d}{x^2 - 1}, \quad (21)$$

$$\left| \log \frac{\mathcal{P}_\lambda}{\mathcal{M}_\lambda} \right| \leq 14d\varepsilon_x + \frac{d}{x^2 - 1} + \frac{H}{L}. \quad (22)$$

More precisely,

$$\left| \log \frac{\mathcal{P}_\lambda}{\mathcal{M}_\lambda} + \frac{H}{L} \right| \leq 14d\varepsilon_x + \frac{d}{x^2 - 1} + \frac{2H_2}{L^2}. \quad (23)$$

Proof. Sum Proposition 2.3 with multiplicities k_r ; the result is $14\varepsilon_x \sum_r rk_r = 14d\varepsilon_x$. Sum Lemma 3.1 to obtain $d/(x^2 - 1)$, and sum (15) to obtain H/L . Finally $a_r \leq 1/L \leq 1/4$, so summing (16) gives

$$0 \leq \log \frac{\mathcal{I}_\lambda}{\mathcal{M}_\lambda} + \frac{H}{L} \leq \frac{2H_2}{L^2}.$$

The triangle inequality proves (23). $\square$

Corollary 4.3 (Asymptotic criterion). *Let λ_y be any sequence of nonempty partitions. If*

$$d_y \varepsilon_{p_y} + \frac{d_y}{p_y^2} \longrightarrow 0, \quad (24)$$

then $\mathcal{P}_{\lambda_y}/\mathcal{J}_{\lambda_y} \rightarrow 1$ and $\mathcal{P}_{\lambda_y}/\mathcal{I}_{\lambda_y} \rightarrow 1$. Under (24),

$$\frac{\mathcal{P}_{\lambda_y}}{\mathcal{M}_{\lambda_y}} \longrightarrow 1 \iff \frac{H_y}{L_y} \longrightarrow 0. \quad (25)$$

Still under (24), the explicit expansion is

$$\log \frac{\mathcal{P}_{\lambda_y}}{\mathcal{M}_{\lambda_y}} = -\frac{H_y}{L_y} + O\left(d_y \varepsilon_{p_y} + \frac{d_y}{p_y^2 - 1} + \frac{H_{2,y}}{L_y^2}\right). \quad (26)$$

Proof. Condition (24) implies $7R_y \varepsilon_{p_y} \leq 7d_y \varepsilon_{p_y} \leq 1/2$ eventually, so Theorem 4.2 applies. The first two conclusions follow from (20)–(21). The refined estimate gives (26). Since $H_2 \leq H$,

$$\frac{2H_2}{L^2} \leq \frac{2}{L} \frac{H}{L}.$$

Thus $H/L \rightarrow 0$ is sufficient. Conversely, with $A_y = 14d_y \varepsilon_{p_y} + d_y/(p_y^2 - 1)$, the refined estimate gives the explicit lower bound

$$\left| \log \frac{\mathcal{P}_{\lambda_y}}{\mathcal{M}_{\lambda_y}} \right| \geq \left(1 - \frac{2}{L_y}\right) \frac{H_y}{L_y} - A_y.$$

Here $A_y \rightarrow 0$, so convergence of the ratio to one forces $H_y/L_y \rightarrow 0$. $\square$

Corollary 4.4 (Uniform families). *For each y , let $\mathfrak{F}_y$ be a nonempty family of finite nonempty partitions, and suppose*

$$D_y = \sup_{\lambda \in \mathfrak{F}_y} d(\lambda) < \infty, \quad H_y^* = \sup_{\lambda \in \mathfrak{F}_y} H(\lambda) < \infty.$$

If

$$D_y \varepsilon_{p_y} + \frac{D_y}{p_y^2} \longrightarrow 0, \quad (27)$$

then

$$\begin{aligned} \sup_{\lambda \in \mathfrak{F}_y} \left| \log \frac{\mathcal{P}_\lambda}{\mathcal{J}_\lambda} \right| &\longrightarrow 0, \\ \sup_{\lambda \in \mathfrak{F}_y} \left| \log \frac{\mathcal{P}_\lambda}{\mathcal{I}_\lambda} \right| &\longrightarrow 0. \end{aligned}$$

Moreover,

$$\sup_{\lambda \in \mathfrak{F}_y} \left| \frac{\mathcal{P}_\lambda}{\mathcal{M}_\lambda} - 1 \right| \longrightarrow 0 \iff \frac{H_y^*}{L_y} \longrightarrow 0. \quad (28)$$

Proof. Since $R(\lambda) \leq d(\lambda) \leq D_y$, condition (27) makes the source smallness hypothesis uniformly valid. Equations (20) and (21) give the first two assertions. Put

$$A_y^* = 14D_y \varepsilon_{p_y} + \frac{D_y}{p_y^2 - 1}.$$

For every $\lambda \in \mathfrak{F}_y$, (23) and $H_2(\lambda) \leq H(\lambda)$ give

$$\left| \log \frac{\mathcal{P}_\lambda}{\mathcal{M}_\lambda} + \frac{H(\lambda)}{L_y} \right| \leq A_y^* + \frac{2}{L_y} \frac{H(\lambda)}{L_y}.$$

The upper bound proves sufficiency. Taking suprema in the corresponding reverse-triangle lower bound gives

$$\sup_{\lambda \in \mathfrak{F}_y} \left| \log \frac{\mathcal{P}_\lambda}{\mathcal{M}_\lambda} \right| \geq \left(1 - \frac{2}{L_y} \right) \frac{H_y^*}{L_y} - A_y^*.$$

Uniform convergence of the positive ratios to one is equivalent to uniform convergence of their logarithms to zero, and necessity follows. $\square$

The quantity $d\varepsilon_x$ is a robust sufficient payment obtained from the available source upper bound. It is not asserted to be logically necessary for the actual, unknown signed prime error.

Corollary 4.5 (Convenient growth hypotheses). *If*

$$\log d_y = o(V(L_y)), \quad H_y = o(L_y), \quad (29)$$

then $\mathcal{P}_{\lambda_y}/\mathcal{M}_{\lambda_y} \rightarrow 1$.

Proof. Because $\log L = o(V(L))$, equation (5) and $\log d = o(V)$ give $d\varepsilon_x \rightarrow 0$. Also $V(L) = o(L)$, so $d/x^2 \rightarrow 0$. Apply Corollary 4.3 and $H = o(L)$. $\square$

Corollary 4.6 (Single-order uniform range). *Let $R_y \geq 1$ be integers. If $\log R_y = o(V(L_y))$, then*

$$\max_{1 \leq r \leq R_y} \left| \log \frac{P_r(y)}{K_r(p_y)} \right| \longrightarrow 0. \quad (30)$$

More sharply, (30) holds whenever, for some fixed $0 < \delta < 0.1853$,

$$R_y \leq \exp((0.1853 - \delta)V(L_y)) \quad (31)$$

for all sufficiently large y .

Proof. For $r \leq R_y$, the three one-factor errors are bounded by

$$14R_y\varepsilon_x + \frac{R_y}{x^2 - 1} + \frac{1}{L}.$$

The first growth hypothesis makes the first two terms vanish. Under (31),

$$R_y\varepsilon_x \leq 0.027L^{1.801}e^{-\delta V(L)} \longrightarrow 0,$$

and $R_y/x^2 \rightarrow 0$ because $V(L) = o(L)$. The source smallness condition is therefore automatic eventually. $\square$

5 Sharpness of the leading-kernel condition

Proposition 5.1 (The all-ones obstruction). *Fix $c > 0$ and let*

$$\lambda_y = 1^{k_y}, \quad k_y = \lfloor cL_y \rfloor.$$

Then

$$\frac{\mathcal{P}_{\lambda_y}}{\mathcal{M}_{\lambda_y}} \longrightarrow e^{-c}. \quad (32)$$

Proof. Here $d_y = H_y = H_{2,y} = k_y$. The source and power errors satisfy $k_y\varepsilon_{p_y} + k_y/p_y^2 \rightarrow 0$, while $k_y/L_y \rightarrow c$ and $k_y/L_y^2 \rightarrow 0$. Equation (23) therefore gives $\log(\mathcal{P}_{\lambda_y}/\mathcal{M}_{\lambda_y}) \rightarrow -c$. $\square$

Thus a degree condition alone cannot guarantee the leading equivalent. The statistic H is not an artifact of the proof: even the simplest growing partition detects the accumulated first correction of the logarithmic integral kernel.

6 Exact artifact, provenance, and claim boundary

The companion artifact has epistemic role `reproduction_not_analytic_proof`. It compiles exactly 96 structured oracle rows:

interface	rows
source constants, endpoint, hazard, and kernel algebra	16
fixed orders $r \in \{1, 2, 3, 4, 8, 16, 32, 64\}$	8
integer partitions of degrees at most 8	66
growth envelopes and sharpness	6
total	96

Every scalar field is checked by an independent exact-type semantic verifier. Twenty-four theorem-interface mutations—including an inclusive endpoint, a missing Stieltjes boundary, a reversed hazard or Jensen inequality, wrong $2r - 1$ factors, deletion of H/L , and a degree-only leading claim—are rejected. Seven separate source-metadata and strict-JSON attacks are also rejected. The canonical certificate has 29,717 bytes and SHA-256

64761d3a85afdee4682982ad545d20a66d2ed69926764bcc9580e0dc8c5f8710.

The immutable Git closure contains 59 exact release blobs: the 51-row RH-384 inherited closure and its eight standard release files. The new Johnston–Yang input is one logical external lock, giving 60 logical source objects. The author-manuscript PDF is not redistributed. Its lock records the versioned URL, DOI, MIME type, 278,380-byte length, 22-page count, and SHA-256, as well as the versioned source-tar and source-`main.tex` hashes. Network verification is disabled

by default and requires the explicit `-network` flag. The publication archive contains the lock record and verifier, but excludes the external PDF and source tar.

The finite regression inherited from RH-384 checks only the degree-eight partition bookkeeping. It neither establishes the Johnston–Yang estimate nor proves any limiting statement. Conversely, the analytic proof above does not depend on a finite numerical fit.

In the surrounding prime-dynamics program, all clocks remain fixed and finite before any prefix limit, and only the universally safe phasewise class with $c_{11} = 0$ is in scope. Nothing in this paper gives a growing clock, an active- c_{11} cancellation theorem, a projectively compatible selector, an effective threshold, an operator or trace formula, a zeta-zero identification, or a proof of the Riemann hypothesis. Gates A–E remain false.

Declarations

Data and code availability. The analytic proof, exact certificate compiler, tests, official closed Draft 2020–12 schema, source-lock records, opt-in verifier, and archive replay tools accompany the paper. The Johnston–Yang PDF and source tar are not redistributed; their exact remote identifiers are recorded in the external lock.

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AI assistance. AI-assisted tools were used for proof-interface enumeration, adversarial testing, manuscript drafting, and release auditing. All mathematical claims, citations, source locks, and final text were reviewed under author responsibility.

References

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- [2] RH research program. Prime-tail scale separation below the quadratic square-clock gap, 2026. Frozen repository release RH-384, used only as a finite regression target.